

8.1 Defining and measuring development

There are a number of different ways to define **development**. Human development generally means the ways in which a country seeks to improve the **standard of living** and **quality of life** for its people. Standard of living refers to the material goods and levels of comfort people have. It can be measured using factors such as income, employment opportunities, and affordable access to healthcare. These factors affect a person's quality of life, which is a more subjective term that can measure factors such as a person's happiness, general wellbeing, and political freedoms, such as their right to vote.

Indicators of development

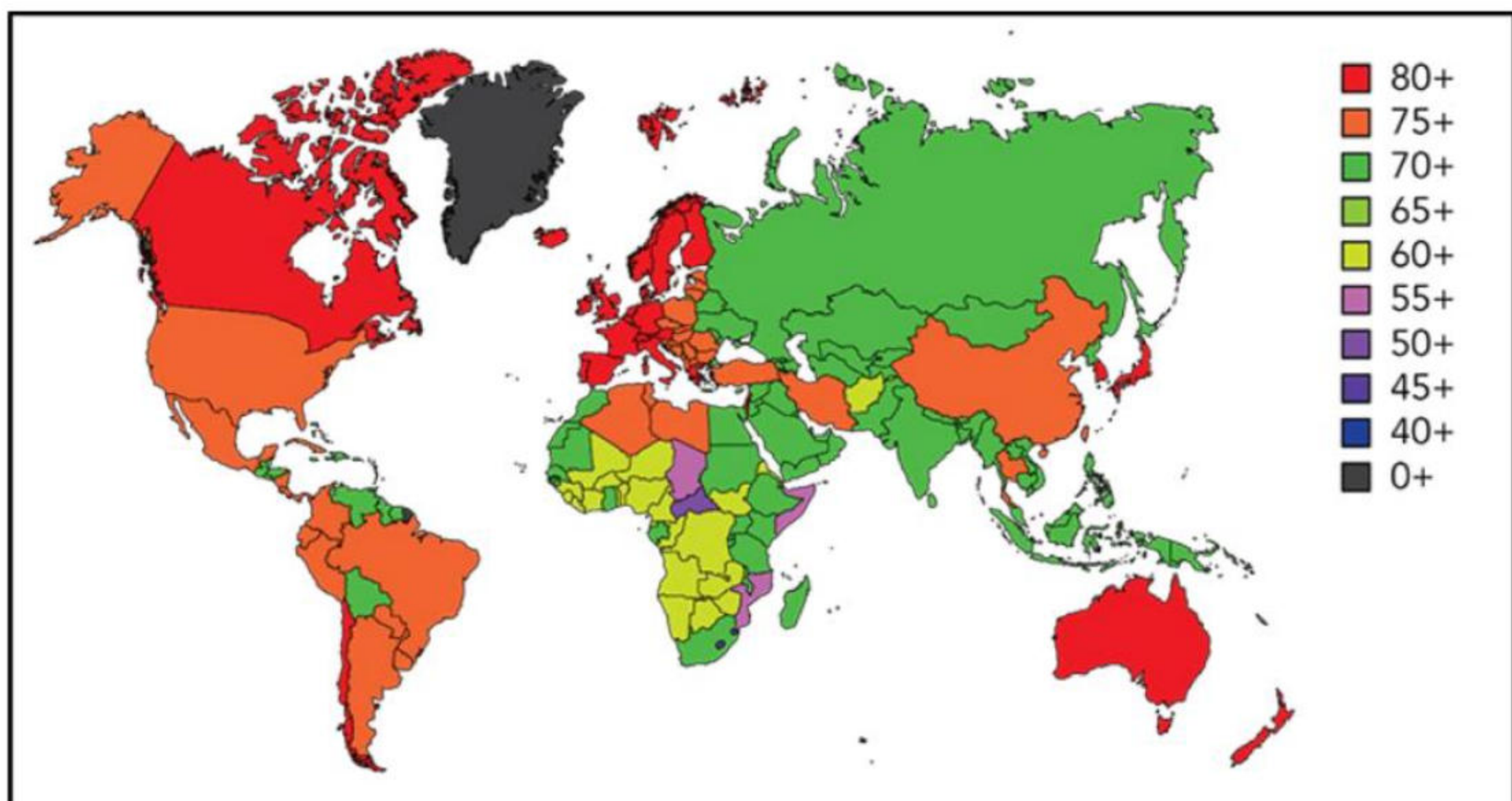
We can measure a country's level of development in different ways. We can use **economic indicators** and **social indicators** to compare countries at different levels of development. Development can also cover political, environmental, or cultural criteria, but Tables 8.1 and 8.2 show the most common economic and social indicators of development.

Table 8.1: Economic measures of development

Gross Domestic Product (GDP)	the total value of goods and services produced in a particular country in a year.
Gross National Product (GNP)	the total value of all the goods and services produced in a particular country (GDP), plus income generated by its residents abroad in a year (it includes the income generated by citizens living abroad but excludes the income earned by foreign residents within the country).
Gross National Income (GNI)	GNI encompasses GDP and GNP. It is the total income earned by a country's people and businesses wherever in the world it is earned.

Table 8.2: Social measures of development

Life expectancy	the average age to which a person can be expected to live.
Infant mortality rate	the number of children who die before reaching the age one, per 1000 live births per year.
Calorie intake	the energy obtained from the food eaten per person each day, measured in kilocalories.
Literacy rate	the percentage of the population over the age of 15 who can read and write.
People per doctor	the ratio of people per doctors, usually expressed per 1000 people.



Source **A**: Map showing life expectancy at birth

TASK 1: Study Source **A**

a Describe the **global pattern** of life expectancy.

- b** Suggest why life expectancy varies across the world.
- c** Do you think life expectancy alone is a good indicator of development in a country? Explain your answer.



Source B: Toilets across the world, from an LIC, MIC, and HIC

TASK 2: Study Source B

- a** Why do you think access to quality toilets could be a good indicator of development?
- b** What other household items could be good indicators of development?
- c** What could be the advantages and disadvantages of using household photographs as indicators of development?
- d** Visit Gapminder's Dollar Street website, which investigates how people live in countries of different stages of development. Select a 'poor' family and a 'rich' family. Explore their homes and everyday items. Now, write a paragraph to describe the similarities and differences between the families you have chosen.

The **Human Development Index (HDI)** is a composite measure of development (Figure 8.1). It considers both economic and social indicators. It provides a summary of global development by combining measures of education, health, and income per capita. These measures are transformed into an index from 0 to 1, with 0 being the lowest level of human development and 1 being the highest.

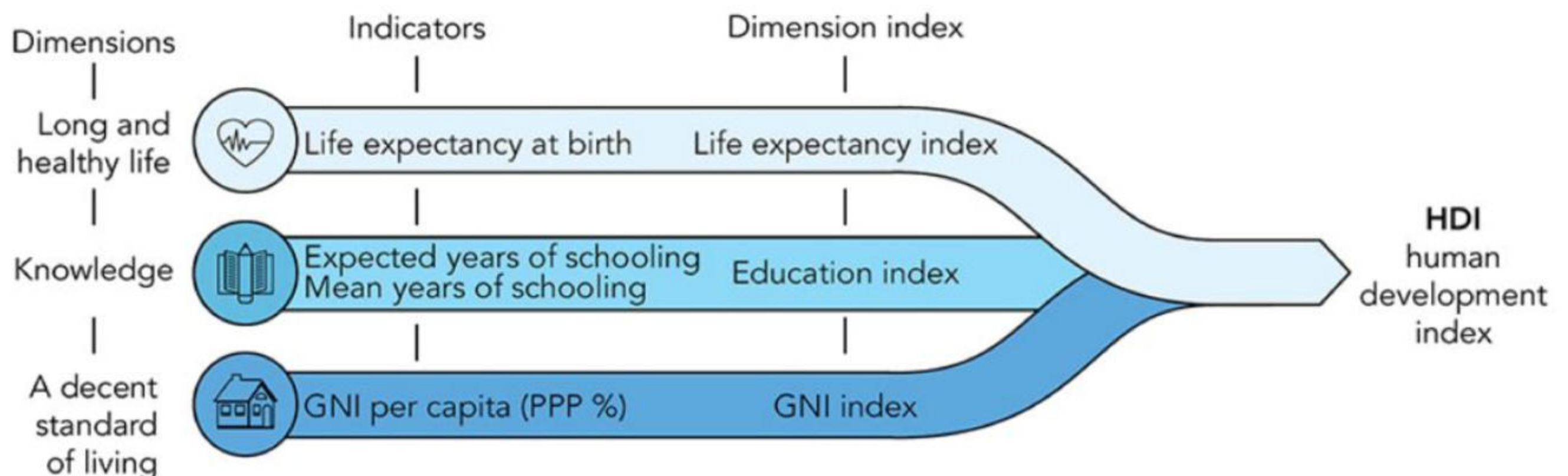
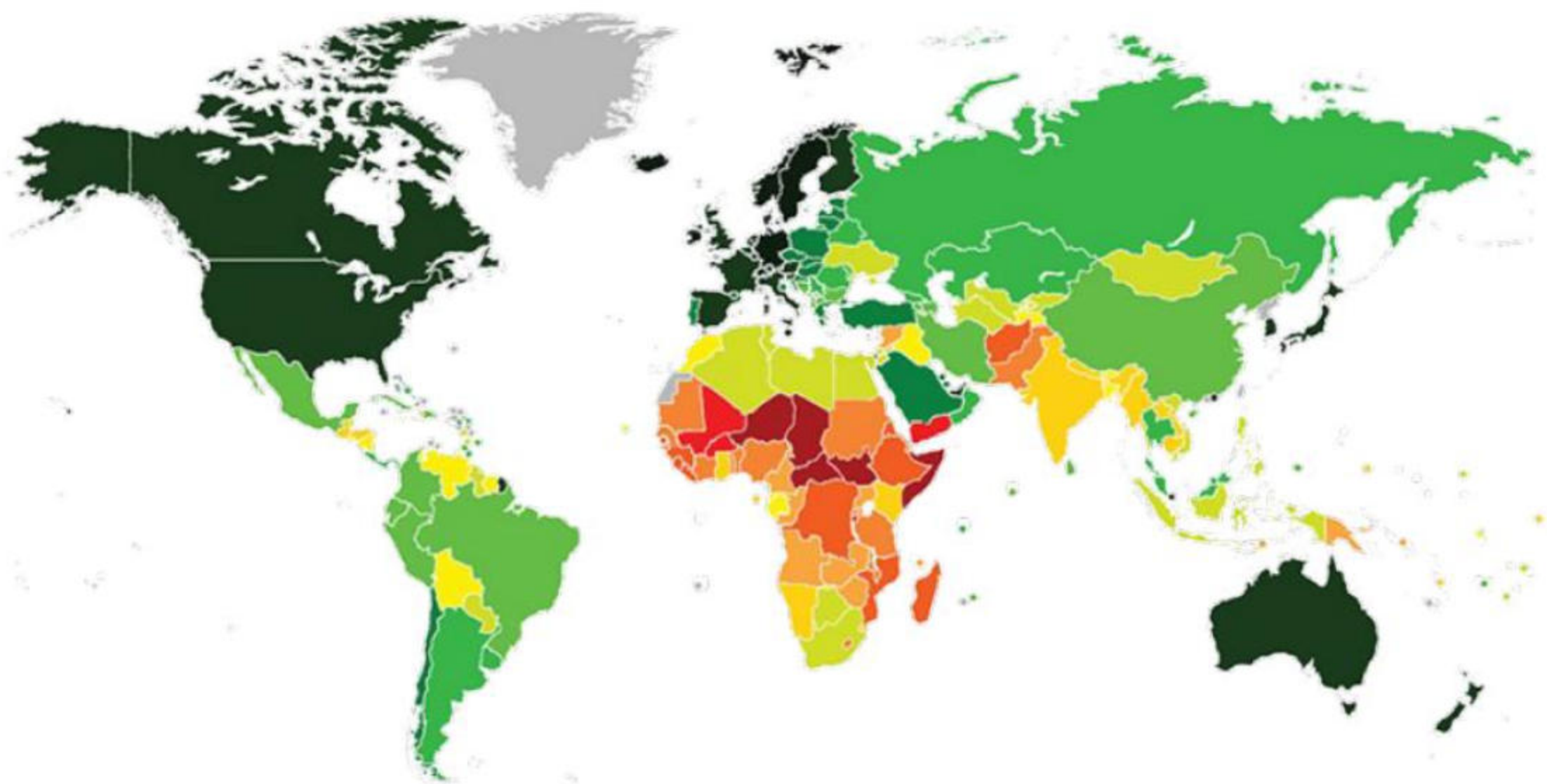


Figure 8.1: The Human Development Index

Table 8.3: UN Human Development data (2021)

Country	HDI value	Life expectancy (years)	Mean years of schooling (years)	Expected years of schooling (years)	GNI per capita (\$PPP)	HDI rank
Switzerland	0.962	84.0	13.9	16.5	66 933	1
Singapore	0.939	82.8	11.9	16.5	90 919	10
USA	0.921	77.2	13.7	16.3	64 765	21
Japan	0.925	84.9	13.4	15.2	42 274	24
Brazil	0.754	72.8	8.1	15.6	14 370	86
Morocco	0.683	74.0	5.9	14.2	7303	122
India	0.633	67.2	6.7	11.9	6590	130
Pakistan	0.544	66.1	4.5	8.7	4624	161
Afghanistan	0.478	62.0	3.0	10.3	1824	177
Chad	0.394	52.5	2.6	8.0	1364	190



World map of countries or territories by Human Development Index scores in increments of 0.050 (based on 2022 data, published in 2024)



Source : World map of countries or territories by Human Development Index

TASK 3: Study Source

- a What are the key indicators used to calculate the Human Development Index?
- b Study Table 8.3. Are there any unexpected results between and/or within countries? Why might this be?
- c Study Source C. **Describe the global pattern** of HDI shown on the map. Suggest reasons for this pattern.
- d In pairs, choose three countries and research their HDI scores over time since 1990 to the present day. **Plot this data** as a line graph. **Describe and explain the trends.**
- e Do you think that HDI or social and economic indicators alone are a better indicator of development? Why?

Evaluating indicators

It is useful to measure and compare how developed places are to study how this changes over time. However, each indicator has advantages and disadvantages, which affect how useful it is as a measure of development. For example, many development indicators are measured at a national scale, and therefore variations between different places and different groups of people are not clear. Development data can also be difficult to reliably collect in some countries and is out-of-date as soon as it is collected.

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Economic indicators

- Indicators, such as GNP, are an average. This means that extreme poverty for the majority of a population, for example, can be skewed by a few wealthy families.
- Recording indicators like GDP reliably means accounting for the earnings of every citizen and business, including informal sector work. This isn't always possible.
- Each country's GDP is converted into US dollars for comparison between countries. This means some data may become unreliable because of changes in currency exchange rates.

Social indicators

- Social indicators provide a much broader understanding of how developed a country is. They are useful in showing how money generated by a country benefits ordinary people.
- Birth rate and death rate, when used on their own, are not reliable indicators because a high birth rate does not necessarily mean lack of development and a low birth rate does not necessarily mean a high level of development.
- Factors such as access to healthcare, the age structure of a country, and government policies can have a big impact on birth rates and death rates.
- In some LICs, birth rates and death rates are not always recorded.
- It is difficult to collect social data in rural areas, informal settlements, and war zones.
- Social indicators are averages across a country and don't always link together, such as health, education, and diet.

Human Development Index

To get a better picture of development in a country, it is better to consider both economic and social indicators together. HDI is a composite measure of development, combining measures of education, health, and income per capita. HDI can be used effectively to show differences between countries and to show global patterns. The United Nations (UN) Development Programme updates the data annually so it is possible to track trends over time.

HDI has disadvantages too. For example, income still has a lot of 'weighting' in the HDI calculation, and therefore HICs regularly score highly. There are also concerns that it is still too simple as it would be better to include other measures such as political and environmental measures.

Table 8.4: Development indicators for selected countries (2022)

Country	HDI	Birth rate	Death rate	Literacy rate
USA	0.927	11/1000	10/1000	86%
Saudi Arabia	0.875	17.5/1000	3/1000	94%
Cuba	0.764	8.9/1000	15/1000	100%
Morocco	0.698	18/1000	6.2/1000	69%
India	0.644	16.4/1000	9/1000	69%

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TASK 4

- a Write your own definitions of development, standard of living, and quality of life. In small groups, discuss your personal definitions.
- b How useful do you think indicators of development are for defining these terms?
- c Carefully study the development data for the selected countries in Table 8.4.
 - i Are there any surprises in the data?
 - ii What does this development data suggest about the problems of using one single development indicator to measure development?
- d Go to The World Bank development data website and choose one economic and one social indicator of development for 15 countries of your choice.

Choose a range of HICs, MICs, and LICs.

- i **Plot a scatter graph** to show the relationship between your chosen economic indicator of development and your chosen social indicator for each country.
- ii **Draw a line of best fit and identify any anomalies.**
- iii Describe and explain your results. Are there any anomalies?